

James A. Michaelov

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Academic Positions

Postdoctoral Researcher: University of Oxford
Department of Experimental Psychology (Faculty Mentor: **James Whittington**)

Postdoctoral Associate: Massachusetts Institute of Technology
Department of Brain and Cognitive Sciences (Faculty Mentor: **Roger Levy**)
MIT Libraries CREOS (Mentor: **Micah Altman**)

Education

Cognitive Science (PhD): University of California San Diego 2024
 Thesis: *Understanding the role of statistics in the predictive processing of language*
 Committee: **Benjamin Bergen**, Seana Coulson, Marta Kutas, Federico Rossano, Victor Ferreira

Cognitive Science (MSc), with Distinction: University of Edinburgh 2018
 Thesis: *Converting from Universal Dependencies to Discourse Representation Theory*
 Supervisor: **Federico Fancellu**

Philosophy and Linguistics (MA Hons), First Class Honours: University of Edinburgh 2017
 Thesis: *How universal are prominence hierarchies? Evidence from native English speakers*
 Supervisors: **Hannah Rohde & Jennifer Culbertson**

Publications

Michaelov, J. A., Amo Alonso, C., Chang, T. A., & Levy, R. P. (in press). Target-Language Generation in Multilingual Models: Activation Steering and Optimal Control. *The 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026)*.

Salhan, S., Arnett, C., **Michaelov, J. A.**, & Buttery, P. (in press). Beetle: A Bilingual Model Suite for Modelling Second-Language Processing. *The 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026)*.

Trott, S., **Michaelov, J. A.** Jones, C. R., Chang, T. A., & Bergen, B. K. (2026). Large Language Models as Distributional Baselines for Language Tasks. *Open Mind*.

Trott, S., Taylor, S. M., Jones, C. R., **Michaelov, J. A.**, & Rivière, P. D. (2026). Language Statistics and False Belief Reasoning: Evidence from 41 Open-Weight LMs. *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026)*.

Michaelov, J. A. & Bergen, B. K. (2026). Better Language Models Better Model the N400, but not Reading Time. *Journal of Memory and Language*, 149, 104762.

Michaelov, J. A., Levy, R. P., & Bergen, B. K. (2025). Language Model Behavioral Phases are Consistent Across Architecture, Training Data, and Scale. *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*.

Michaelov, J. A., Estacio, R., Zhang, Z., & Bergen, B. K. (2025). Not quite Sherlock Holmes: Language model predictions do not reliably differentiate impossible from improbable events. *Findings of the Association for Computational Linguistics: ACL 2025*, 13528-13551.

- Arnett, C., Chang, T. A., **Michaelov, J. A.**, & Bergen, B. K. (2025). On the Acquisition of Shared Grammatical Representations in Bilingual Language Models. *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025)*, 20707-20726.
- Michaelov, J. A.**, Arnett, C., & Bergen, B. K. (2024). Revenge of the Fallen? Recurrent Models Match Transformers at Predicting Human Language Comprehension Metrics. *The Proceedings of the First Conference on Language Modeling (COLM)*.
- Michaelov, J. A.** & Bergen, B. K. (2024). On the Mathematical Relationship Between Contextual Probability and N400 Amplitude. *Open Mind*, 8, 859–897.
- Michaelov, J. A.**, Bardolph, M. D., Van Petten, C. K., Bergen, B. K., & Coulson, S. (2024). Strong Prediction: Language model surprisal explains multiple N400 effects. *Neurobiology of Language*, 5(1), 107–135.
- Michaelov, J. A.**, Arnett, C., Chang, T. A., & Bergen, B. K. (2023). Structural priming demonstrates abstract grammatical representations in multilingual language models. *The 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP 2023)*, 3703–3720.
- Michaelov, J. A.** & Bergen, B. K. (2023). Emergent inabilities? Inverse scaling over the course of pretraining. *Findings of the Association for Computational Linguistics: EMNLP 2023*, 14607–14615.
- Michaelov, J. A.** & Bergen, B. K. (2023). Ignoring the alternatives: The N400 is sensitive to stimulus preactivation alone. *Cortex*, 168, 82–101.
- Michaelov, J. A.** & Bergen, B. K. (2023). *Rarely* a problem? Language models exhibit inverse scaling in their predictions following *few*-type quantifiers. *Findings of the Association for Computational Linguistics: ACL 2023*, 14162–14174.
- Rezaii, N., **Michaelov, J. A.**, Josephy-Hernandez, S., Ren, B., Hochberg, D., Quimby, M., & Dickerson, B. C. (2023). Measuring Sentence Information via Surprisal: Theoretical and Clinical Implications in Nonfluent Aphasia. *Annals of Neurology*, 94, 647–657.
- Trott, S., Jones, C. R., Chang, T. A., **Michaelov, J. A.**, & Bergen, B. K. (2023). Do Large Language Models know what humans know? *Cognitive Science*, 47(7), e13309.
- Michaelov, J. A.** & Bergen, B. K. (2022). Collateral facilitation in humans and language models. *Proceedings of the 26th Conference on Computational Natural Language Learning (CoNLL 2022)*, 13–26.
- Michaelov, J. A.** & Bergen, B. K. (2022). Do language models make human-like predictions about the coreferents of Italian anaphoric zero pronouns?. *Proceedings of the 28th International Conference on Computational Linguistics (COLING 2022)*, 1–14.
- Michaelov, J. A.**, Coulson, S., & Bergen, B. K. (2022). So Cloze yet so Far: N400 Amplitude is Better Predicted by Distributional Information than Human Predictability Judgements. *IEEE Transactions on Cognitive and Developmental Systems*, 15(3), 1033–1042.
- Michaelov, J. A.** & Bergen, B. K. (2020). How well does surprisal explain N400 amplitude under different experimental conditions?. *Proceedings of the 24th Conference on Computational Natural Language Learning (CoNLL2020)*, 652–663.
- Michaelov, J. A.** (2017). The Young and the Old: (T) Release in Elderspeak. *Lifespans and Styles* 3(1), 2–9.

Preprints

- Michaelov, J. A.**, Arnett, C., Chang, T. A., Rivière, P. D., Taylor, S. M., Jones, C. R., Trott, S., Levy, R. P., Bergen, B. K., & Altman, M. (2026). How Open Must Language Models be to Enable Reliable Scientific Inference?. *arXiv:2603.26539*.

Non-Archival Refereed Conference Papers

Michaelov, J. A. & Levy, R. P. (2026). N-gram-like Language Models Predict Reading Time Best. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 48. Rio de Janeiro, Brazil.

Michaelov, J. A. & Arnett, C. (2025). Disaggregation Reveals Hidden Training Dynamics: The Case of Agreement Attraction. *First Workshop on CogInterp: Interpreting Cognition in Deep Learning Models*. San Diego, USA.

Michaelov, J. A., Coulson, S., & Bergen, B. K. (2023). Can Peanuts Fall in Love with Distributional Semantics?. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 45. Sydney, Australia.

Michaelov, J. A. & Bergen, B. K. (2022). The more human-like the language model, the more surprisal is the best predictor of N400 amplitude. *NeurIPS 2022 Workshop on Information-Theoretic Principles in Cognitive Systems (InfoCog)*. New Orleans, USA. (**Spotlight**)

Jones, C. R., Chang, T. A., Coulson, S., **Michaelov, J. A.**, Trott, S., & Bergen, B. (2022). Distributional Semantics Still Can't Account for Affordances. In *Proceedings of the Annual Meeting of the Cognitive Science Society*, 44. Toronto, Canada.

Michaelov, J. A., Bardolph, M. D., Coulson, S., & Bergen, B. K. (2021). Different kinds of cognitive plausibility: why are transformers better than RNNs at predicting N400 amplitude?. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 43. Vienna, Austria.

Conference Presentations and Posters

Michaelov, J. A. & Bergen, B. K. (2026). Better Language Models Better Model the N400, but not Reading Time. *The 18th Annual Meeting of the Society for the Neurobiology of Language (SNL 2026)*. Geneva, Switzerland.

Salhan, S., Arnett, C., **Michaelov, J. A.**, & Buttery, P. (2026). Beetle: A Bilingual Model Suite for Modelling Second-Language Processing. *Crosslinguistic Perspectives on Processing and Learning (X-PPL 2026)*. Bayonne, France.

Michaelov, J. A., & Levy, R. P. (2026). 'On the observed scaling effects in language model predictions of reading time'. *The 39th Annual Conference on Human Sentence Processing (HSP 2026)*. Cambridge, USA.

Michaelov, J. A., & Levy, R. P. (2025). 'The effect of orthographic neighborhood density on reading time in 9 languages'. *The 38th Annual Conference on Human Sentence Processing (HSP 2025)*. College Park, USA.

Rezaei, N., **Michaelov, J. A.**, Josephy-Hernandez, S., Ren, B., Quimby, M., Hochberg, D., & Dickerson, B. C. (2023). 'Measuring Sentence Information via Surprisal: Clinical and Theoretical Implications in Nonfluent Aphasia'. *Society for Neuroscience Meeting (SfN 2023)*. Washington, D.C., USA.

Arnett, C., Chang, T. A., **Michaelov, J. A.**, & Bergen, B. K. (2023). Crosslingual Structural Priming and the Pre-Training Dynamics of Bilingual Language Models. *The 3rd Multilingual Representation Learning Workshop (MRL 2023)*. Singapore.

Michaelov, J. A., Coulson, S., & Bergen, B. K. (2022). 'Do we need situation models? Distributional semantics can explain how peanuts fall in love'. *The 35th Annual Conference on Human Sentence Processing (HSP 2022)*. Santa Cruz, USA.

Michaelov, J. A., Coulson, S., & Bergen, B. K. (2022). 'Cloze behind: Language model surprisal predicts N400 amplitude better than cloze'. *The 35th Annual Conference on Human Sentence Processing (HSP 2022)*. Santa Cruz, USA.

Michaelov, J. A., Bardolph, M. D., Coulson, S., & Bergen, B. K. (2021). 'Is the relationship between word probability and processing difficulty linear or logarithmic?'. *The 34th CUNY Conference on Human Sentence Processing (CUNY 2021)*. Philadelphia, USA.

Michaelov, J. A., Bardolph, M. D., Coulson, S., & Bergen, B. K. (2020). ‘Surprisal is a good predictor of the N400 effect, but not for semantic relations’. *AMLaP 2020: 26th Architectures and Mechanisms for Language Processing conference*. Presentation in *Special Session: Computational models of language processing*. Potsdam, Germany.

Michaelov, J. A., Culbertson, J., & Rohde, H. (2017). ‘How universal are prominence hierarchies? Evidence from native English speakers’. *AMLaP 2017: 23rd Architectures and Mechanisms for Language Processing conference*. Lancaster, UK.

Michaelov, J. A. (2017). ‘The Young and the Old: (t) Release in Elderspeak’. *ULAB 2017: 7th Undergraduate Linguistics Association of Britain Conference*, University of Cambridge, Cambridge, U.K.

Media Interviews

Sandrine Ceurstemont (2023). Bigger, Not Necessarily Better. *Communications of the ACM*.
(Interviewed and quoted in article).

Invited Talks and Panels

What do language models learn when they learn language?	
<i>Language and Intelligence Lab</i> , Massachusetts Institute of Technology.	2026
<i>Natural Language Processing Seminar</i> , Cornell Tech, Cornell University.	2026
What can psycholinguistics and natural language processing learn from each other?	
<i>Language & Cognition Research Seminar</i> , Harvard University.	2025
<i>Computation and Psycholinguistics Lab</i> , New York University.	2025
<i>Cog Lunch</i> , Massachusetts Institute of Technology.	2025
Panelist for ‘Decoding the Mind: What is thought?’	
<i>Ideas on Consciousness</i> , University of California, Berkeley.	2025
What can language models tell us about the N400?	
<i>Center for Research in Language</i> , University of California San Diego.	2024
<i>Distinguished Speakers in Language Science</i> , Saarland University.	2024
<i>Psycholinguistics Group</i> , Saarland University.	2024
Using language models to understand the N400 (and vice-versa)	
<i>Computational Psycholinguistics Laboratory</i> , Massachusetts Institute of Technology.	2023
<i>Language Processing Group</i> , University of California, Irvine.	2023
Can Peanuts Fall in Love with Distributional Semantics?	
<i>Lupyan Lab</i> , University of Wisconsin-Madison.	2023
<i>Cognition at the Shore</i> , University of California San Diego.	2023
Panelist for ‘Academic Panel: Postgraduate Studies’	
<i>The 8th Undergraduate Linguistics Association of Britain Conference</i> , The University of Edinburgh.	2018

Student Mentorship

University of California San Diego	
Reeka Estacio: <i>Undergraduate Research Assistant Mentee, Co-author</i>	2023-2025
Zhien Zhang: <i>Undergraduate Research Assistant Mentee, Co-author</i>	2023-2025
Van Nguyen: <i>Honors Program Graduate Student Advisee</i>	2024
Norah Kerendian: <i>Undergraduate Research Assistant Mentee</i>	2023
Srija Sankavaram: <i>Undergraduate Research Assistant Mentee</i>	2023

Awards, Grants, and Fellowships

Honorable Mention, School of Social Science Dissertation of the Year, UCSD	2025
Cognitive Science Diversity Teaching Assistant Award: for promoting accessibility, diversity of perspective, and an inclusive culture as a Teaching Assistant in the Department of Cognitive Science at the University of California San Diego.	2023
Cognitive Science Society Student Travel Grant (\$1,200)	2023
CARTA Annette Merle-Smith Fellowship (\$20,000)	2021-2022
Center for Academic Research and Training in Anthropogeny Fellowship (\$20,000)	2020-2021
Glushko Travel and Research Account (\$500/year)	2018-2022

Teaching Experience

Teaching Assistant: University of California San Diego

<i>Introduction to Python</i>	2024 (Spring Quarter)
<i>Data Science in Practice</i>	2024 (Winter Quarter)
<i>Introduction to Data Science</i>	2023 (Fall Quarter)
<i>Learning, Memory, and Attention</i>	2023 (Spring Quarter)
<i>Neurobiology of Cognition</i>	2023 (Winter Quarter)
<i>Cognitive Consequences of Technology</i>	2022 (Fall Quarter)
<i>Cognitive Perspectives</i>	2022 (Summer Session 2)
<i>What the *#!?: An Uncensored Introduction to Language</i>	2021 (Fall Quarter)
<i>Cognitive Neuroeconomics</i>	2020 (Fall Quarter)
<i>Cognitive Neuroeconomics</i>	2020 (Summer Session 2)
<i>Language Comprehension</i>	2020 (Summer Session 1)
<i>Cognitive Neuroeconomics</i>	2020 (Winter Quarter)
<i>What the *#!?: An Uncensored Introduction to Language</i>	2019 (Fall Quarter)
<i>Minds and Brains</i>	2019 (Spring Quarter)

Tutor (Teaching Assistant): University of Edinburgh

<i>Logic 1</i>	2018 (Semester 2)
<i>Informatics 1: Computation and Logic</i>	2017 (Semester 1)

Peer Review Experience

Journals:

Journal of Memory and Language: Reviewer
Language, Cognition and Neuroscience: Reviewer
Northern European Journal of Language Technology: Reviewer
Lifespans and Styles: Reviewer

Conferences:

The Annual Meeting of the Cognitive Science Society (CogSci): Reviewer, Meta-Reviewer
The Annual Meeting of the Association for Computational Linguistics (ACL): Reviewer, Area Chair
The Conference on Computational Natural Language Learning (CoNLL): Area Chair
The Annual Conference on Neural Information Processing Systems (NeurIPS): Reviewer
The Conference on Language Modeling (COLM): Reviewer
The Workshop on Cognitive Modeling and Computational Linguistics (CMCL): Reviewer
The Workshop on Interpreting Cognition in Deep Learning Models (CogInterp): Reviewer

Service Roles

Program Committee: Cognitive Modeling and Computational Linguistics Workshop	2023–2025
Graduate Student Representative: Department of Cognitive Science, UCSD	2019–2020
Cognitive Science Department Representative: UCSD Graduate Student Association	2018–2019
National Committee: The Undergraduate Linguistics Association of Britain	2017–2018
Webmaster: The University of Edinburgh Linguistics and English Language Society	2016–2017
Webmaster: The Undergraduate Linguistics Association of Britain	2015–2016
Treasurer: The University of Edinburgh Linguistics and English Language Society	2015–2016

Other Professional Experience

Applied Scientist Intern: Amazon (Alexa Games), *Seattle, WA* 2023
Project: Utilizing and developing state-of-the-art natural language understanding systems.
Skills: Applied research, production-level code, product development, product evaluation.

Tools and Resources Developed

Parallel architecture (Parc) Language Models:

<https://huggingface.co/collections/jmichaelov/parc-models-68111455eee2f8461b8b593e>

Language models of different architectures trained in parallel to facilitate analyses of differences between model architectures and their training dynamics. Currently includes Pythia, Mamba-1, and RWKV-4 models trained on the same dataset with the same 6 random seeds for data shuffling and model initialization, with intermediate training checkpoints provided.

PsychFormers:

<https://github.com/jmichaelov/PsychFormers>

Command-line tool that allows the user to use transformer neural network language models to calculate metrics that are relevant to psycholinguistic experiments. Uses the *transformers* Python package and the OpenAI GPT-3 API.

Easy-FAS:

<https://github.com/jmichaelov/easy-fas>

Command-line tool that allows a user to easily calculate the Forward Association Strength between any two words based on norms gathered from human participants. Uses the *Edinburgh Associative Thesaurus* and the *University of South Florida Free Association Norms*.

Word Embedding Similarity Calculator:

<https://github.com/jmichaelov/word-embedding-similarity>

Command-line tool that allows the user to calculate the similarity between two words, or the similarity between a word and its context, based on word embeddings. Compatible with *GloVe* and *fastText*.

Contributions to Large Open Source Projects

Language Model Evaluation Harness:

<https://github.com/EleutherAI/lm-evaluation-harness>

Framework for the standardized testing of generative language models on a large number of different evaluation tasks. Widely used in research and as the backend for benchmark leaderboards.

Regular Contributor (see <https://zenodo.org/records/18636344>)